

PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule.
PROCEDURE: Not answered.
SPECIFIC CLINICAL QUESTION: Not answered.
OUTSIDE TISSUE DIAGNOSIS: Not answered.
PRIOR MALIGNANCY: Not answered.
CHEMORADIATION: Not answered.
ORGAN TRANSPLANT: Not answered.
IMMUNOSUPPRESSION: Not answered.
OTHER DISEASES: Not answered.

UUID: EDF3F3EB-AC62-403D-9180-147C2CFD2AAS
TCGA-BJ-A023-01A-PR



100-0-3
carcinoma, papillary, follicular variant
8340/3

ADDENDA:

Addendum

Molecular Anatomic Pathology Testing:

Site: thyroid, nos c73.9

hw
11/27/11

Part 1:

- A. **BRAF V600E mutation IDENTIFIED at low level (see note).**
B. Mutations in **NRAS61, HRAS61, KRAS12/13, and RET/PTC1, RET/PTC3 and PAX8/PPARg** rearrangements NOT identified.

Note:

The BRAF mutation was identified and confirmed by direct sequencing. However, low levels of mutant DNA were seen despite quite pure population of epithelial cells tested (based on the ratio of *GAPDH* and *KRT7* controls), which indicates that a small fraction of tumor cells is positive for this mutation. The prognostic importance of BRAF mutation when present in a small fraction of tumor cells is not known and may be less significant as compared to a more typical situation when this mutation is found at high level.

Nucleic acids were extracted in the amount sufficient for testing. Amplification of *GAPDH* and *KRT7* controls was acceptable. Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF V600E* () mutation has been associated with more aggressive behavior of papillary carcinoma (2) and may be used for preoperative risk stratification of patients with papillary thyroid cancer (3). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (4). Based on a recent prospective study of FNA samples (5) and our experience at , detection of *BRAF* mutations and *RET/PTC* and *PAX8/PPARg* rearrangements in thyroid FNA samples correlated with ~100% probability of cancer, whereas detection of *RAS* mutation correlated with malignancy in 83-87% of cases and with benign follicular adenoma or hyperplastic nodule in 13-17%. However, single cases of false-positive *BRAF* mutation detection in thyroid FNA samples have been reported in the literature (6) and therefore the results of molecular testing should be interpreted in combination with clinical information, imaging, and cytology.

- Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006, 30:216-222
- Elisei R, et al. *BRAF V600E* mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008, 93:3943-3949.
- Xing M, et al. *BRAF* mutation testing of thyroid fine-needle aspiration biopsy specimens for preoperative risk stratification in papillary thyroid cancer. *J Clin Oncology* 2010, 27:2977-82.
- Nikiforova MN, et al. *RAS* point mutations and *PAX8-PPARg* rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003, 88: 2318-26.
- Nikiforov YE, et al. Molecular Testing for Mutations in Improving the Fine Needle Aspiration Diagnosis of Thyroid Nodules. *J Clin Endocrinol Metab*. 2009 Mar 24. [Epub ahead of print]
- Chung KW, et al. Detection of *BRAF V600E* mutation on fine needle aspiration specimens of thyroid nodule refines cyto-pathology diagnosis, especially in *BRAF V600E* mutation-prevalent area. *Clin Endocrinol (Oxf)*. 2006, 65(5):660-6.

Sample Preparation and Procedure

Thyroid FNA samples were collected in the nucleic acid preservative solution. For cytology samples and frozen samples, extraction of DNA and RNA was performed from the sample provided using standard laboratory procedure. Optical density readings were obtained. Real-time PCR was performed on the i platform to amplify *BRAF*, *NRAS* codon 61, *HRAS* codon 61, and *KRAS* codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of *RET/PTC1*, *RET/PTC3*, and *PAX8-PPARg* rearrangements on ABI7500. If required, the mutation type was confirmed by Sanger sequencing of the PCR products on . Amplification of the *GAPDH* and *KRT7* genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. DNA or RNA from samples positive for each of these mutations were used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10% of alleles with mutation present in the background of normal DNA and RNA.

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
RNA Discrepancy		
Prior Malignancy/History		
Dual/Synchronous Primary Nodule		
Cytology (circle):		
Reviewer Initials	Date Reviewed: 11/27/11	

FINAL DIAGNOSIS:

PART 1: THYROID, TOTAL THYROIDECTOMY (29 GRAMS) ✓

- A. PAPILLARY THYROID CARCINOMAS, TWO LEFT LOBE FOCI (1.3 AND 0.8 CM), FOLLICULAR VARIANTS, CONFINED TO THYROID; NO ANGIOLYMPHATIC INVASION. ✓
- B. CHRONIC LYMPHOCYTIC THYROIDITIS.
- C. ONE ISTHMIC PERITHYROIDAL LYMPH NODE, NO TUMOR PRESENT (0/1).
- D. PATHOLOGIC STAGE: pT1 N0.

PART 2: LYMPH NODES, CENTRAL COMPARTMENT, SELECTIVE DISSECTION – TWO LYMPH NODES WITH NO TUMOR PRESENT (0/2).

COMMENT:

Molecular studies are pending and will be reported in an addendum.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Total Thyroidectomy
TUMOR SITE:	Left Lobe ✓
TUMOR FOCALITY:	Multifocal ✓
TUMOR SIZE (largest nodule):	Greatest Dimension: 1.3 cm
HISTOLOGIC TYPE:	Papillary carcinoma, follicular variant
PATHOLOGIC STAGING (pTNM):	pT1
	pN0
	Number of regional lymph nodes examined: 3
	Number of regional lymph nodes involved: 0
	pMX

MARGINS:

Margins uninvolved by carcinoma

VENOUS/LYMPHATIC (LARGE/SMALL VESSEL) INVASION (V/L):

Absent

ADDITIONAL PATHOLOGIC FINDINGS:

Thyroiditis